C Supplementary analysis

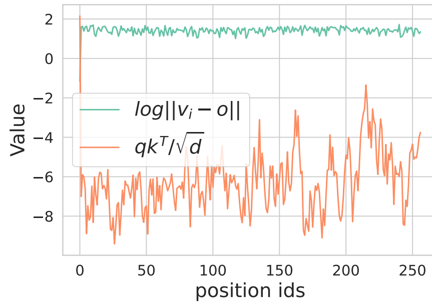

Figure 10 The range of fluctuation of $\log|v_i - o|$ and $\frac{qk_i^T}{\sqrt{d}}$ in a single decoding step. Compared to $\frac{qk_i^T}{\sqrt{d}}$, $\log|v_i - o|$ is stable, hence we do not consider $\log|v_i - o|$ in our proposed sampling probability.

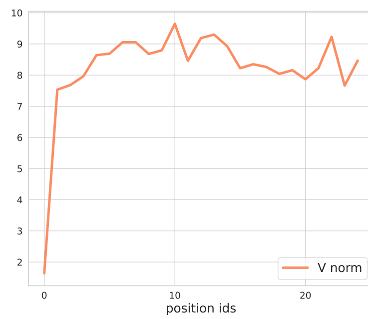

Figure 11 The y-axis is the norm of values states $\|v_i\|$ for token i (on the x-axis). We observe that the value norm $\|v_0\|$ of the attention sink is significantly smaller than others.

Figure 10 shows that compared to $\frac{qk_i^T}{\sqrt{d}}$, $\log|v_i - o|$ is stable in a decoding step. Figure 11 shows that the norm of the value states of attention sink is smaller than others.

D Additional evaluation

In this section, we provide additional experimental results to demonstrate that

- MagicPIG can support longer context lengths and a wide range of LLMs (Appendix D.1).
- MagicPIG can scale up with 70B level LLM (Appendix D.2).
- MagicPIG can perform well in reasoning benchmarks (Appendix D.3).
- MagicPIG improves decoding throughput with various hyper-parameters (K, L). (Appendix D.4).

D.1 Longer Contexts

Following the setups of Table 3, we evaluate two additional models, MegaBeam-Mistral-7B-512K[4] and Llama3-8B-Prolong-512K (Gao et al., 2024) with context lengths extended to 256K. The results are shown in Table 4.

Table 4 Synthesized tasks on RULER (Hsieh et al., 2024). MagicPIG preserves high accuracy with extended context lengths and different models. Config and cost are defined as in Table 1.

Methods	Config	16K	32K	64K	96K	128K	256K	Avg.	Cost_1	Cost_2	Cost_{total}.
MegaBeam-Mistral-7B-512K	Full	91.7	88.1	83.5	83.7	83.5	82.5	85.5	0.00	1.00	1.00
MagicPIG	(10,150)	89.8	86.5	81.7	80.7	81.6	79.0	83.2	0.00	0.02	0.02
MagicPIG	(9,120)	90.7	88.5	82.9	82.4	82.3	80.1	**84.5**	0.00	0.04	0.04
MagicPIG	(8,75)	90.6	86.4	82.8	81.6	82.3	80.8	84.1	0.00	0.05	0.05
Quest	(16,0.04)	83.3	83.2	79.3	78.6	78.5	78.5	80.2	0.06	0.04	0.10
Llama3-8B-Prolong-512K	Full	93.5	90.8	85.1	83.5	81.7	78.4	85.5	0.00	1.00	1.00
MagicPIG	(10,150)	88.0	86.4	81.3	78.8	77.3	71.1	80.5	0.00	0.02	0.02
MagicPIG	(10,170)	89.0	88.7	82.8	80.0	77.7	73.7	82.0	0.00	0.025	0.025
MagicPIG	(9,120)	91.4	88.2	82.4	80.4	79.2	75.2	**82.8**	0.00	0.04	0.04
MagicPIG	(8,75)	91.4	88.6	83.1	80.5	79.1	73.9	82.8	0.00	0.05	0.05
Quest	(16,0.04)	84.9	83.7	78.7	78.6	76.3	72.3	79.2	0.06	0.04	0.10

D.2 Scaling up to larger models

We evaluate MagicPIG for meta-llama/Llama-3.1-70B-Instruct (Dubey et al., 2024) to demonstrate that our approach can work well with larger LLMs in Table 5.

Table 5 Synthesized tasks from RULER (Hsieh et al., 2024). MagicPIG preserves high accuracy with low computation for 70B level models. 4 layers {0,16,32,48} are preserved. Config and cost are defined as in Table 1.

Methods	Config	16K	32K	64K	96K	Avg.	Cost_1	Cost_2	Cost_{total}.
Llama-3.1-70B-Instruct	Full	96.4	94.6	89.2	80.8	90.3	0.00	1.00	1.00
MagicPIG	(10,150)	94.7	93.5	87.5	79.3	88.8	0.00	0.02	0.02
MagicPIG	(9,110)	95.7	93.5	88.4	79.4	89.3	0.00	0.034	0.034
MagicPIG	(9,120)	95.5	94.1	88.8	80.6	89.8	0.00	0.04	0.04

D.3 Reasoning

In mathematical reasoning tasks infini_igsm (Zhou, 2024a,b), MagicPIG consistently outperforms Quest (Tang et al., 2024) across all complexity (in terms of operators). We also find TopK attention suffers from significant performance degradation while Oracle Sampling can maintain high accuracy.

[4] https://huggingface.co/aws-prototyping/MegaBeam-Mistral-7B-512k

Table 6 Tasks from infini_igsm (Zhou, 2024a,b). MagicPIG preserves high accuracy for reasoning tasks. Config and cost for MagicPIG and Quest are defined as in Table 1. Config denotes the ratio of selected tokens for TopK and sampled tokens for oracle sampling. For oracle sampling, massive duplication exists in sampled tokens, so Cost_2 is significantly lower than the ratio of sampled tokens Theorem 3.3.

Task	Methods	Config	2-Ops	4-Ops	5-Ops	Cost_1	Cost_2	Cost_{total}.
4K close (Zhou, 2024a)	*Llama-3.1-8B-Instruct*	Full	87.4	71.4	26.8	0.00	1.00	1.00
	MagicPIG	(10,300)	83.1	67.2	20.7	0.00	0.06	0.06
	MagicPIG	(10,220)	79.8	58.9	17.9	0.00	0.04	0.04
	MagicPIG	(10,150)	68.3	43.5	11.7	0.00	0.02	0.02
	TopK	0.06	78.6	62.9	20.8	0.50	0.06	0.56
	TopK	0.04	76.2	59.0	19.2	0.50	0.04	0.54
	TopK	0.02	71.5	44.0	11.3	0.50	0.02	0.52
	Oracle Sampling	0.3	88.1	72.4	27.6	0.50	0.02	0.52
	Oracle Sampling	0.1	88.5	69.2	26.2	0.50	0.01	0.51
	Oracle Sampling	0.02	83.1	57.9	11.9	0.50	0.005	0.505
	Quest	(16,0.06)	55.8	23.2	5.2	0.06	0.06	0.12
8K close (Zhou, 2024b)	*Llama-3.1-8B-Instruct*	Full	80.2	68.8	26.0	0.00	1.00	1.00
	MagicPIG	(10,300)	78.6	61.5	25.2	0.00	0.06	0.06
	MagicPIG	(10,220)	72.2	60.7	20.4	0.00	0.04	0.04
	MagicPIG	(10,150)	67.1	44.0	11.9	0.00	0.02	0.02
	TopK	0.06	70.2	61.1	22.3	0.50	0.06	0.56
	TopK	0.04	66.9	55.2	20.6	0.50	0.04	0.54
	TopK	0.02	64.7	47.2	15.9	0.50	0.02	0.52
	Oracle Sampling	0.3	80.0	67.3	26.2	0.50	0.02	0.52
	Oracle Sampling	0.1	76.6	64.1	25.4	0.50	0.01	0.51
	Oracle Sampling	0.02	79.0	60.3	20.4	0.50	0.005	0.505
	Quest	(16,0.06)	54.8	30.0	11.1	0.06	0.06	0.12

Table 7 System performance for MagicPIG using Llama-3.1-8B-Instruct with a 96K context length under varying hyper-parameter configurations. We report the decoding latency (time between tokens, TBT) when the batch size is 1, the maximum throughput, and the throughput with a latency constraint of 200ms ($\text{Throughput}_{200\text{ms}}$ in the table). Config and cost are defined as in Table 1. The number with * means hit the memory limit of CPU.

Config	TBT (ms)	Max Throughput (tokens/sec)	$\text{Throughput}_{200\text{ms}}$ (tokens/sec)	Cost_{total}.
(11,300)	17.38	41.68*	40.84	0.02
(10,220)	14.07	32.29*	26.66	0.04
(10,170)	16.79	46.52*	39.90	0.025
(10,150)	18.31	53.78	48.89	0.02
(9,120)	13.93	32.50	26.60	0.04
(8,75)	12.47	27.43	21.17	0.05

D.4 System performance

In this section, we evaluate the system performance (latency, throughput) of MagicPIG under different hyper-parameter configurations. We use Llama-3.1-8B-Instruct (Dubey et al., 2024) with 96K contexts as an example.

E Selection of hyper-parameter (K, L)

In this section, we discuss the impact of the LSH hyper-parameter (K, L) and how to select it. First, we briefly explain what hyper-parameter (K, L) does for LSH sampling. Then, we explain the relations between (K, L) and attention computation cost and accuracy. Finally, we show how we decide the parameters by ablation studies.